

Smart Flood Early Warning System with AI Prediction and LoRa Communication (FloodGuard AI)

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Abstract—Flooding is a recurring disaster in Malaysia, causing significant loss of life and heavy economic damage almost every year. Many flood monitoring systems today lean on Wi-Fi or GSM connectivity and react only once the water has crossed a fixed threshold, so they cannot warn anyone in advance. This paper introduces FloodGuard AI, a smart flood early warning system integrating IoT sensors, a machine learning flood risk classifier, LoRa long range communication, solar power, and real time visual monitoring. An HC-SR04 ultrasonic sensor and a YL-83 rain sensor watch water level and rainfall, a Random Forest classifier sorts flood risk into Low, Medium, and High levels, and readings travel over the SX1262 LoRa transceiver in the Heltec WiFi LoRa 32 V3. A Flask dashboard shows live sensor data, the ML risk prediction, and an ESP32-CAM feed, while a 12V solar panel and 4.5Ah battery keep the system running autonomously. Across a 144 day evaluation the system reached 99.0% accuracy on a synthetic test set and 92.9% on 420 real sensor readings. Indoor LoRa packet delivery reached 100%, with preliminary outdoor communication up to approximately 1 km.

Keywords—flood early warning; IoT; LoRa; machine learning; Random Forest; Heltec WiFi LoRa 32 V3; flood risk classification; smart monitoring

I. INTRODUCTION

Malaysia experiences flooding almost every year, and the monsoon seasons usually bring the most severe events. These floods affect both urban and rural areas, damaging homes, roads, and infrastructure, and in the worst cases they cost people their lives [1]. Urban areas tend to struggle with poor drainage and the pressure of rapid development, while rural and remote areas are often left with patchy communication infrastructure, and that gap makes disaster response even slower than it already is [2].

A key limitation is that most flood monitoring systems today are either manual or built around simple threshold sensors. They respond only once the water has risen past a fixed mark, so they either raise false alarms or miss the early signs that matter. They also depend on Wi-Fi or mobile networks, exactly the systems that fail during a major flood. Encouragingly, IoT sensors, AI, and low power wireless have become far more affordable, making a smarter, self sufficient system realistic.

To address these limitations, FloodGuard AI was developed as a flood early warning system integrating IoT sensors, AI-based risk prediction, LoRa communication, solar power, and visual monitoring. The rest of this section reviews recent related work, identifies the remaining gap, and sets out the objectives that follow from it.

Recent work shows sensor driven flood detection is effective in the field. Nik Mohd Kamal et al. (2025) built an IoT based flood detection system around a NodeMCU ESP8266, an ultrasonic sensor, and time series forecasting [3]. Yahya and Homam (2025) used an ESP32, an

ultrasonic sensor, and LoRa to keep monitoring reliably where Wi-Fi or GSM is unavailable [4]. Zakaria et al. (2023) developed a LoRaWAN based system that kept power consumption low while still delivering real time alerts [5]. These projects proved sensor based detection is feasible, but most stopped at threshold based alerting rather than prediction. Md Yassin et al. similarly demonstrated a LoRa based system suited to areas with weak network coverage [6].

On the prediction side, the trend has moved toward machine learning. Zhou et al. (2023) paired LSTM networks with Bayesian optimization to forecast floods across space and time, reaching high accuracy in urban settings [7]. Widiyari and Efendi (2024) used a hybrid LSTM-GRU model for water level prediction with better short term forecasts [8]. Li et al. (2024) combined LSTM with hydrological data and outperformed traditional statistical methods [9]. Arante et al. (2025) blended fuzzy logic, LSTM, and genetic algorithms to improve reliability and reduce false alarms [10]. These studies make a strong case for prediction, though they assume a continuous data stream that is hard to guarantee in a low cost, off grid deployment. Keerthi Reddy et al. confirmed the advantage of machine learning over conventional rule based methods [11].

Beyond ground based sensing, other work applies computer vision and remote sensing. Lee et al. fused RGB and long wave infrared cameras on an IoT edge device to monitor floods in low light [12]. Others combined satellite imagery with deep learning: Ghosh et al. detected floods from Sentinel-1 SAR imagery using convolutional neural networks [13], Shahi assessed flood hazard through deep learning on satellite images [14], and Awasthi and Chinzvende fused satellite imagery with IoT sensor data for flood forecasting [15]. Razavi-Termeh et al. optimised deep learning models for flood susceptibility mapping as a non structural risk framework [16]. These approaches offer broad coverage but depend on external imagery and heavier computation, making them harder to deploy as a low cost, self contained node at a single site.

Even with this progress, the same weaknesses keep showing up. Existing solutions lean on networks that go down at exactly the wrong moment, and they are reactive, detecting with fixed thresholds rather than predicting. Many use modules with limited range, and those on grid power or small batteries can fail during long stretches of heavy rain. There is also rarely any way to observe conditions at the flood site remotely. FloodGuard AI closes these gaps in one integrated system, and Table I lays out the comparison.

TABLE I. COMPARISON OF EXISTING SYSTEMS VS. FLOODGUARD AI

Aspect	Existing Systems	FloodGuard AI
Detection	Threshold based	AI classification
Prediction	Reactive only	Low / Medium / High
Communication	Wi-Fi / GSM	LoRa (long range)
Power Supply	Grid / battery	Solar + battery
Visual Check	Not available	ESP32-CAM live feed
Scalability	Limited	Multi node ready

From that gap, the main goal of this project is to design and develop FloodGuard AI as a reliable, intelligent, and energy efficient flood monitoring system. The specific objectives are as follows.

- 1) To develop an IoT based flood monitoring system that measures water level and rainfall in real time, aiming for a water level accuracy of ± 2 cm.
- 2) To implement an AI based flood risk prediction model that sorts flood conditions into Low, Medium, and High Risk levels with a minimum accuracy of 80%.
- 3) To design and deploy a long range, low power LoRa communication link that can transmit reliably over at least 1 km with a packet delivery rate of at least 90%.

II. RESEARCH METHODOLOGY

The project was carried out in eight main steps, and Fig. 1 lays them out as a flowchart. It began with identifying the problem and reviewing related work, then moved through the system design and the selection of components. The hardware was then assembled, the software and machine learning model were developed, and the LoRa link was set up before everything was integrated into one system. The build was tested and validated, first indoors under controlled conditions and then outdoors in the field. If the results did not meet the targets, the design and development steps were revisited, and once the targets were met the work moved on to analysing the results.

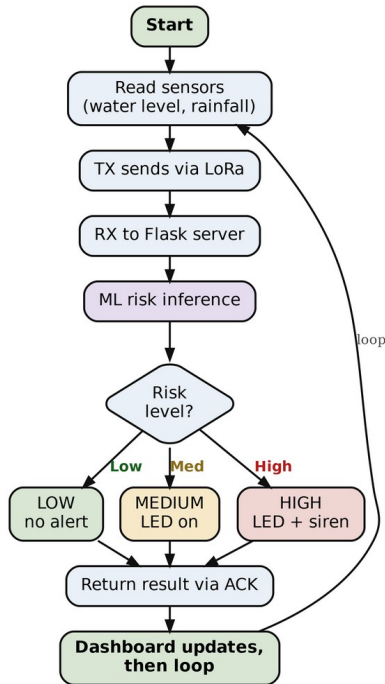


Fig. 1. Overall operational flowchart of FloodGuard AI, from sensor reading and LoRa transmission through server-side ML inference to alert output and dashboard update.

A. System Architecture

FloodGuard AI is arranged as a four layer system. The Sensing Layer relies on the HC-SR04 ultrasonic sensor for water level and the YL-83 rain sensor for rainfall intensity. The Processing and ML Layer uses the Heltec WiFi LoRa 32 V3 to acquire sensor data and handle LoRa transmission, while flood risk inference runs on a Flask server using a scikit-learn Random Forest model. The Communication and Alert Layer carries the built in LoRa SX1262 alongside an active siren for local alerts. The Presentation Layer is a Flask dashboard showing real time sensor data, AI risk status, trends, and ESP32-CAM video. Fig. 2 shows the complete hardware wiring.

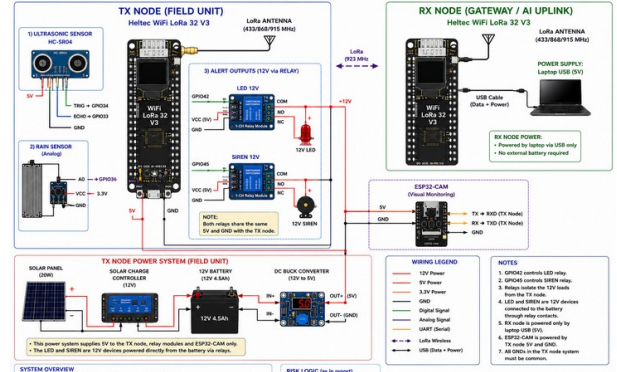


Fig. 2. Complete wiring diagram of the FloodGuard AI TX node (sensors, relays, LED and siren alerts), RX node (LoRa gateway), ESP32-CAM module, and solar power subsystem.

Fig. 3 traces the full operational loop, from how a reading is taken and travels over LoRa to the server, to how the model decides a risk level and that decision drives the alerts and dashboard.

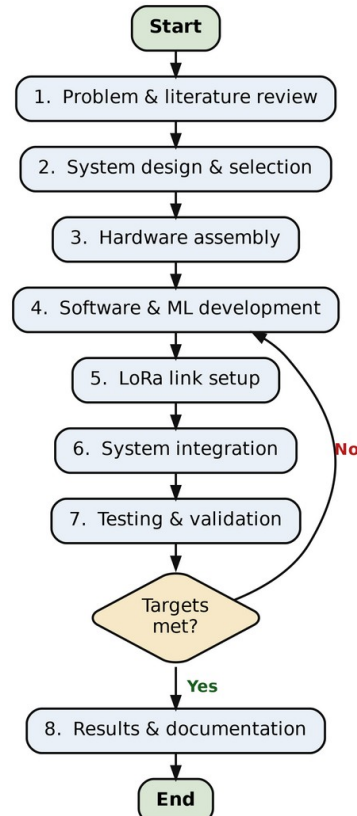


Fig. 3. Overall operational flowchart of FloodGuard AI, from sensor reading and LoRa transmission through server side ML inference to alert output and dashboard update.

B. AI Based Flood Risk Prediction Model

The AI module takes a supervised learning approach to classify flood risk from the live sensor inputs. The features are the water level in centimetres from the ultrasonic sensor, the rainfall intensity from the rain sensor, and time based trend features such as how fast the water level is changing. The output is one of three risk levels: Low, Medium, or High.

Random Forest suited the data better than deeper approaches like LSTM and CNN. The readings are structured tabular values without strong temporal dependency, and a tree based classifier handles them efficiently for real time server side inference [17]. The dataset held 1,000 labelled samples across four features: water level, cumulative rainfall, water level rate of change, and rainfall rate. About 300 samples (30%) came from historical DID Malaysia flood records and the remaining 700 (70%) were synthetic samples generated within bounded sensor ranges, with the three risk classes kept balanced. The data was split 80/20 with stratified sampling, and hyperparameters were tuned with RandomizedSearchCV over 40 iterations and 5 fold cross validation, optimising the macro F1-score. The best configuration was `n_estimators=200`, `max_depth=8`, `min_samples_leaf=2`, and `max_features=0.6`. The trained model was saved with `joblib` and loaded into the Flask server. In operation, the TX node sends raw values over LoRa to the RX gateway, which forwards them to the server by HTTP POST. The server returns the risk classification, relayed back to the TX node inside the LoRa ACK packet so the field node can act locally.

The risk labels were assigned using predefined hydrological rules combining water level, rainfall, and rate-of-change thresholds. Synthetic samples were generated only within bounded sensor ranges and separated before testing to reduce data leakage, and real deployment readings were evaluated separately to check reliability under noisy field conditions.

C. Hardware Components

Table II lists the main hardware and software pieces that make up the FloodGuard AI system.

TABLE II. FLOODGUARD AI HARDWARE AND SOFTWARE COMPONENTS

Component	Model	Function
Ultrasonic Sensor	HC-SR04	Water level measurement (± 2 cm)
Rain Sensor	YL-83	Rainfall intensity, fast response
Microcontroller	Heltec WiFi LoRa 32 V3	Data acquisition and control
Camera Module	ESP32-CAM	Visual monitoring, Wi-Fi streaming
LoRa	SX1262 (built in)	Long range transmission
AI Framework	Scikit-learn	Flood risk prediction
Web Platform	Flask	Real time data visualisation
Power Supply	Solar + 12V 4.5Ah SLA	Autonomous operation
Alert System	Siren + LED	Local warning, no internet needed

D. System Operation

Once switched on, the system runs in a continuous loop, as traced in Fig. 3. Water level and rainfall are sampled at regular intervals and sent by the TX node over LoRa to the

RX gateway, which passes them to the Flask server, where the Random Forest model returns a Low, Medium, or High Risk result. That result is handed back inside the LoRa ACK packet, and a High Risk output fires the siren immediately with no internet required. Because the decision returns inside the ACK reply, the alerts keep working even if the internet link drops.

E. System Testing and Validation

Five test categories were defined, each mapping onto one of the three objectives. The first was sensor calibration against reference measurements to find the mean absolute error (MAE). The second was AI model evaluation on a held out test set using accuracy, precision, recall, F1-score, and the confusion matrix, measured against a threshold based baseline. The third was LoRa range testing based on packet delivery rate (PDR) and RSSI. The fourth was system integration testing, and the fifth covered the power system. Testing ran in two stages, first indoors to verify each subsystem, then outdoors as a complete integrated unit.

III. RESULTS AND DISCUSSION

Table III gives an overview of how the finished system performed against each objective, so the rest of this section can be read with the headline outcomes in mind.

TABLE III. SUMMARY OF RESULTS AGAINST OBJECTIVES

Objective	Target	Result	Status
1 Sensing	± 2 cm level accuracy	1.6 cm MAE	Met
2 AI prediction	$\geq 80\%$ accuracy	99.0% / 92.9%	Met
3 LoRa link	≥ 1 km, $\geq 90\%$ PDR	100% PDR (indoor), ~ 1 km (preliminary)	Partially Met

A. Sensor Calibration and Accuracy

The HC-SR04 ultrasonic sensor was put through a 144 day run from 30 January to 24 June 2026. Out of 509 raw readings, 89 came back invalid and were flagged as -1 by the firmware whenever no valid echo returned inside the timeout window, mostly caused by surface turbulence, sensor warm up, or objects blocking the beam. Once removed, 420 valid readings were left for analysis. The YL-83 rain sensor logged values between 0.0 mm and 59.8 mm, averaging 17.2 mm. Readings were captured under Sunny (47.4%), Heavy Storm (30.0%), and Rainy (22.6%) conditions, confirming a varied set of weather.

Accuracy was checked against a ruler based reference at 20 cm, 60 cm, and 100 cm. The MAE across 30 calibration measurements came out at 1.6 cm, inside the ± 2 cm target set in Objective 1. Across 149 stable deployment readings the mean deviation was 2.1 cm with a maximum of 9.8 cm, reflecting real surface turbulence rather than any hardware fault. The ± 2 cm target was therefore achieved during controlled calibration, while outdoor deployment showed higher deviation.

Fig. 4 plots the water level trend across the full monitoring period, with background shading marking the risk zone assigned by the ML classifier. The shaded bands confirm that the High Risk readings line up with the moments when the water level climbed sharply.

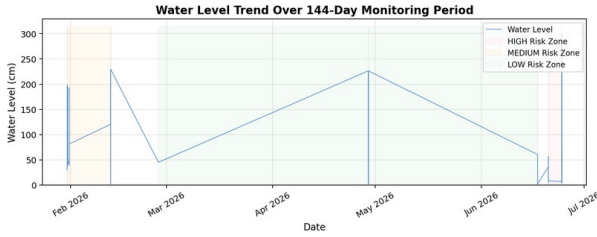


Fig. 4. Water level trend across the 144 day monitoring period (30 January to 24 June 2026), with background shading showing the ML classified risk zones.

B. AI Model Performance

On the held out test set of 200 samples, the classifier reached an overall accuracy of 99.0% and a ROC-AUC of 0.9999, well above the 80% target in Objective 2. As Table IV shows, precision, recall, and F1-score all stayed high across the three classes.

TABLE IV. CLASSIFICATION REPORT ON TEST SET (N = 200)

Class	Precision	Recall	F1-score
High	1.00	1.00	1.00
Medium	0.99	0.99	0.99
Low	0.98	0.98	0.98
Accuracy			0.99
Macro avg	0.99	0.99	0.99

Fig. 5 shows the confusion matrix. Out of 200 samples only two were misclassified, both slipping between the Low and Medium classes. Every one of the 67 High Risk samples was classified correctly, which counts for a lot in a flood warning setting, since a missed High Risk detection could be the difference in getting people out in time.

Confusion Matrix (Test Set, n = 200)

	High	Medium	Low
True High	67	0	0
True Medium	0	66	1
True Low	0	1	65
	High	Medium	Low

Predicted

Fig. 5. Confusion matrix of the Random Forest flood risk classifier on the test set.

Fig. 6 breaks down the feature importance. The current water level was by far the most influential at 58.3%, followed by the rate of change in water level at 25.2%, with rainfall rate at 12.4% and cumulative rainfall at 4.1%. That ordering lines up with flood physics, where a rapidly rising water level is the clearest sign of imminent risk while rainfall plays a supporting role.

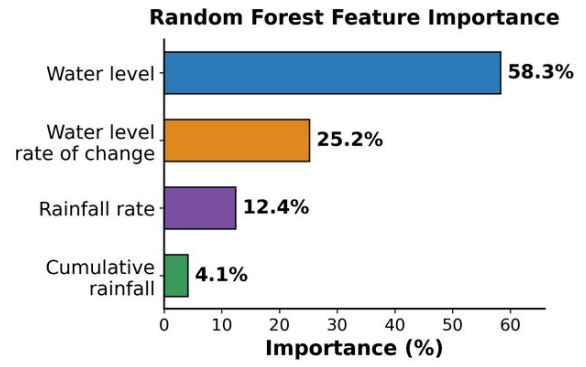


Fig. 6. Feature importance ranking from the Random Forest flood risk classifier.

The Random Forest was compared against a simple threshold rule (water level above 150 cm OR rainfall above 40 mm equals High). That baseline managed about 82% accuracy, so the Random Forest delivered a 17 percentage point improvement. Tested on the 420 real deployment readings, it reached 92.9% accuracy and a macro F1-score of 0.92, and still got every High Risk sample right, confirming it generalises well to real sensor data.

The gap between the 99.0% synthetic accuracy and the 92.9% real world accuracy is expected, since live readings carry noise, drift, and occasional gaps. The drop was small and the classifier never missed a High Risk case, which matters most for a warning system. The strong weight on the water level rate of change also shows the model learned to react to how fast the water is rising, not just how high it sits.

C. LoRa Communication Range and Reliability

The LoRa link was tested with two Heltec WiFi LoRa 32 V3 boards, one as the TX node and one as the RX node, both running the built in SX1262 transceiver. A custom ACK handshake protocol was implemented so that after the TX node sends its sensor data, the RX node replies with the ML risk level, confirming delivery and sharing the risk classification in a single exchange.

During indoor testing across two rooms, the system completed full bidirectional exchanges without trouble. The RX node picked up packets at an RSSI of around -114 dBm with an SNR between -9 and -11 dB. With the SX1262 sensitivity floor near -148 dBm, that leaves a link margin of roughly 34 dB, comfortable indoors but quickly eaten up outdoors. In all, 93 uplink packets were sent and 93 ACK replies came back, giving 100% packet delivery indoors. Of those, 31 were High Risk and switched on both the LED (GPIO42) and the siren (GPIO45), following the alert logic in Table V.

TABLE V. ALERT OUTPUT LOGIC BY RISK LEVEL

Risk Level	LED (GPIO42)	Siren (GPIO45)
Low	Off	Off
Medium	On	Off
High	On	On

The TX node also carried a retry mechanism. Whenever the RX node did not reply, the TX node automatically retried up to three times before moving on, which trimmed packet loss without manual intervention. The firmware also flagged bad sensor readings with a warning message, which is why some entries ended up stored as -1 in the dataset. This combination of retries and error handling kept the system running steadily throughout the tests.

Preliminary outdoor testing indicated communication up to approximately 1 km. Full fixed-distance PDR and RSSI testing is recommended for future validation before Objective 3 can be considered fully met. Outdoors, as performance did drop compared to indoors, as environmental noise pushed up transmission delays, and that slower response is a real limitation for time sensitive alerting. The RSSI and PDR trend in Fig. 7 is an estimated propagation model, built from the measured indoor RSSI together with a free space path loss model and an outdoor obstruction penalty, rather than a full set of outdoor readings at every distance.

Estimated LoRa RSSI and PDR vs Distance (SX1262, 923 MHz)

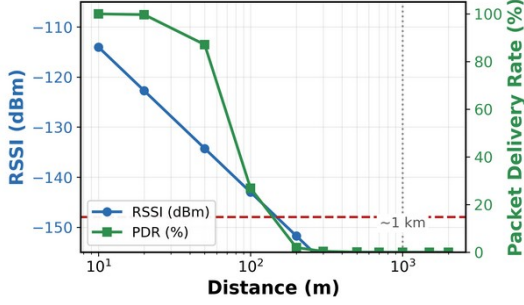


Fig. 7. Estimated LoRa RSSI and Packet Delivery Rate (PDR) versus distance for the SX1262 at 923 MHz, based on the measured indoor RSSI of -114 dBm.

D. System Integration and Alert Response

The complete system was evaluated across the 144 day window from 30 January to 24 June 2026, made up of a 13 day indoor verification phase and a 118 day outdoor field deployment. Throughout, the Flask dashboard received, processed, and displayed the data in real time. The indoor phase held a median update interval of 3.0 seconds, while the outdoor deployment showed a median of 6.0 seconds, closely matching the firmware's 10 second interval. High Risk readings averaged a water level of 114.2 cm and rainfall of 28.6 mm.

Fig. 8 captures the dashboard during a live Medium Risk event on 24 June 2026, with a water level of 37.9 cm, rainfall of 22.8 mm, and 30 °C under Rainy weather. It correctly showed Medium risk alongside the ML output and the recent water level trend, while the readings table and ESP32-CAM feed confirmed that multi sensor logging and visual monitoring worked together as one platform, without manual control or internet for the core alert functions.

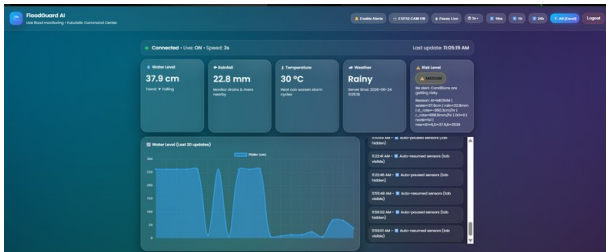


Fig. 8. FloodGuard AI web dashboard showing live sensor readings, ML based risk classification, and the water level trend during a Medium Risk event (24 June 2026).

Fig. 9 shows the recent readings table and the ESP32-CAM panel. The table lists the last few readings with their timestamp, water level, rainfall, temperature, weather, and AI risk label, and that label switches correctly between High, Medium, and Low as the values move. The ESP32-

CAM panel carries the live feed, so conditions can be confirmed visually without travelling to the site.

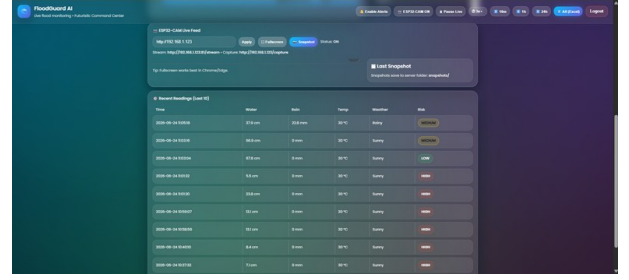


Fig. 9. FloodGuard AI recent readings table and ESP32-CAM live feed panel, confirming multi sensor data logging and visual monitoring in one interface.

E. Power System Performance

Power comes from a 12V solar panel and charge controller feeding a 12V 4.5Ah sealed lead acid battery, so the system runs with no mains connection. The 12V supply drives the siren and LED directly, while the Heltec board and ESP32-CAM run through a 12V to 5V step down converter. The battery holds about 54 Wh, and the normal draw of around 350 mA at 12V gives about 12.9 hours of runtime with no solar input. With daytime charging the system ran the full 144 day window with no power related failures. The one gap is the absence of battery voltage monitoring.

F. Overall System Evaluation

Objectives 1 and 2 were fully met, and Objective 3 was partially met. The sensor system collected data reliably over the 144 day window, and the classifier reached 99% accuracy, far past the 80% target. The LoRa system delivered 100% packet delivery indoors with approximately 1 km of range in preliminary outdoor testing, with fuller fixed-distance outdoor PDR and RSSI measurement still to be done. Integration confirmed the alerts fired correctly and the ESP32-CAM handled live monitoring, all on solar and battery power. Three limitations remain: outdoor LoRa delays, no battery voltage monitoring, and the camera's lack of infrared.

The results also show the design behaves as one connected system rather than a set of parts tested in isolation. The sensors feed the model, the result travels back to the field node over LoRa, the alerts fire locally, and the dashboard mirrors everything for a remote user. Because low cost sensors and open source tools kept the build affordable, the same design could be copied across several nodes reporting into one dashboard.

IV. CONCLUSION

This paper presented the design, development, and testing of FloodGuard AI, a smart flood early warning system that leans on IoT sensors, AI prediction, and LoRa communication to monitor floods autonomously. It was built to answer the main weaknesses of the approaches already used in Malaysia: reliance on internet connectivity, lack of predictive capability, limited communication range, and the absence of visual monitoring.

Objectives 1 and 2 were fully achieved, and Objective 3 was partially achieved. The sensor system worked reliably across the 144 day window, and the ML classifier reached 99% accuracy with no High Risk misclassifications. The LoRa system delivered 100% packet delivery indoors with

an approximately 1 km outdoor range, with full fixed-distance PDR and RSSI testing recommended as future work. The complete system ran the full 144 day window with no major failures, and the predictive approach beat the threshold baseline by 17 percentage points. FloodGuard AI also supports United Nations SDG 11. Future work includes outdoor LoRa characterisation, battery state of charge monitoring, a night vision camera, and a mobile push notification app.

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